A New Quantization Scheme for Classical Games*

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Abstract. We study one special type of quantum games, which we call "games with quantum decisions". The concept of these game goes back to the very beginning of the history of stufying quantum games — namely to the work of Eisert, Wilkens and Lewenstein [8]. We present a new scheme for games quantization which roughly repeats one proposed by these authors, but can be applied for far more wide range of games. We also describe one interesting example of such application for the election process, where quantization scheme leads voters to a better common benefit than they would achieve in classical voting.

Keywords: Measure, automata

1 Introduction

Study of quantum games have been developing since late 1990s, following the works [16, 11, 8]. Perhaps this interest could be explained by the relative affordability of these games for real physical experiments.

In recent years especially many efforts have been made to the study of nonlocal games — such as famous CHSH [4] and Mermin-Ardehali [10, 1] games. In these games players are provided with some settings and must declare their decisions in the form of common classical information.

In this paper we try to revive interest in earliest ideas about quantizing games. In these games, unlike nonlocal ones, players are allowed to declare their decisions by providing a quantum system in some state, i.e. in the form of quantum information. Since this class of games seems not to have any conventional name so far, we proposed to call them "games with quantum decisions".

One of the most widely studied models of game quantization was initially proposed in [8] and further developed in a number of works, up to recent physical implementation on a quantum computer [5]. In this model the game process follows the predefined protocol, which consists of the following steps:

1. a referee prepares a multipartite quantum system (in an entangled state);
2. parts of this quantum system are being distributed among the players;

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3. players perform arbitrary operations on their particles (by applying some unitary operators on their states);
4. the referee collects all the system together, applies some additional unitary operations and performs a measurement;
5. payoffs are being paid to the players according to the results of the measurement.

This is a general description of the protocol, which needs one significant remark. The fact is that when one builds a quantum version of a classical game, he might intend to keep the quantum game analogous to its classical version. In order to do so, he must properly define operations made by the referee.

Traditionally this model is being applied for quantization of non-cooperative games (such as Prisoner's Dilemma [8] or Kolkata Restaurant Problem [15]), since they are comparatively easy analyzable. But it is also applicable for quantization of cooperative games, in which entanglement effects could affect also the process of forming coalitions.

Initially this model was demonstrated on the quantization of Prisoner's Dilemma (see Section 2 for details), which is a 2×2 matrix game. The one-dimensional space of the referee's operations

$$\widehat{J} \in \left\{ \exp\left(i\gamma \begin{pmatrix} 0001 \\ 0010 \\ 0100 \\ 1000 \end{pmatrix} \right) : \gamma \in \left[0, \frac{\pi}{4}\right] \right\} \tag{1}$$

(where parameter γ represents the amount of the initial entanglement and thus the power of the quantum effect in the game) was chosen in order to keep the same outcome of the game for (a) all four pure classical strategies and (b) all two-dimensional space of mixed strategies in the mixed expansion of the original Prisoner's Dilemma.

This model is very naturally applicable to a game with any number of players N, just by substituting the $\begin{pmatrix} 0001 \\ 0010 \\ 0100 \\ 1000 \end{pmatrix}$ with $2^N \times 2^N$ exchange matrix in (1) (e.g. similar technique was introduced in [3]). On the other hand, this model has quite specific limitation on the amount of strategies for each player. Namely, each player in the classical version of a game should have exactly 2 strategies (and respectively one-dimensional space of strategies in the mixed expansion of this game, where the single parameter represents the probability of choosing one or another of the two options).

Probably the first step towards an M-choice generalization of an N-player game is done in [15] by constructing a quantum minority game, where three players use three-level quantum states.

In this work we introduce rather new approach to the generalizing similar scheme to the set of arbitrary multiplayer multichoice game.

1.1 Outline

In Section 2 of the work we start with describing the initial idea of [8] and repeat its application for Prisoner's Dilemma. We use slightly different but obviously

equivalent notations, in order to emphasize the general similarity of the initial idea with our subsequent approaches.

Section 3 (which actually is the main contribution of this work) is dedicated to formal description of the new quantization scheme, and in Section 4 we illustrate this technique on a specific example of election process. The key idea is repeated in both sections in order to keep them self contained to a certain degree, and to allow the reader to skip or just to run over one of them.

Finally, in Section 5 we discuss some known issues and comments on game quantization.

2 Prisoner's Dilemma

Prisoner's Dilemma is one of the most popular non-zero-sum games. It is considered that the initial source of the idea of this game was discussions by Merrill Flood and Melvin Dresher in 1950, as part of the game theory investigations in the Rand Corporation during the Cold War. As Flood and Dresher did not publicize their ideas in external journal articles, there is no canonical formulation of the rules of this game. Let us quote the website `www.prisoners-dilemma.com`, which is dedicated entirely to the Prisoner's Dilemma:

> You and a friend have committed a crime and have been caught. You are being held in separate cells. You are both offered a deal but have to decide what to do. But you are not allowed to communicate with your partner and you will not be told what they have decided until you have made a decision.
> Essentially the deal is this.
>
> – If you confess and your partner denies taking part in the crime, you go free and your partner goes to prison for five years.
>
> – If your partner confesses and you deny participating in the crime, you go to prison for five years and your partner goes free.
>
> – If you both confess you will serve four years each.
>
> – If you both deny taking part in the crime, you both go to prison for two years.

These rules could be represented as in Table 1.

This game is a famous illustration of the fact that two rationally operating individuals sometimes might not cooperate, even if it is in their interest to do so. The phenomenon represented by this game became very popular among scientists of different areas, and it was investigated in many works and from many points of view. For the history of this idea we refer to one of the most significant books on this topic, which is dedicated entirely to the Prisoner's Dilemma [13].

Table 1. Players' payoffs in Prisoner's Dilemma

		Your partner: Confesses	Your partner: Denies
You	Confess	4 years for You, 4 years for Your partner	0 years for You, 5 years for Your partner
	Deny	5 years for You, 0 years for Your partner	2 years for You, 2 years for Your partner

2.1 Classical version

Analysis of such games usually assumes the Nash equilibrium as a solution concept [1]. Thus fairly simple arguments in the case of the Prisoner's Dilemma show that rational players will confess and serve four years each.

Let us formalize the game process in the following way (some redundant formalism here will be clarified later):

1. The game board consists of two numbers a and b from $[0,1]$.
   Initially their values are $a_1 = b_1 = 0$.

2. Players independently choose (say linear) operators A and B and apply them to a and b, respectively.
   Let us denote $a_2 = A(a_1)$ and $b_2 = B(b_1)$.

3. Resulting numbers a_2 and b_2 are treated as probabilities of choosing the second strategy, i.e., denying participation in the crime. (Obviously, $1 - a_2$ and $1 - b_2$ then will be the probabilities of choosing the first strategy, i.e., confessing.)

4. Players get their payoffs — respectively
   $\$_1(a_2, b_2) = -4(1-a_2)(1-b_2) - 5a_2(1-b_2) - 2a_2b_2 = 4b_2 - 4 - a_2(1+b_2)$
   and
   $\$_2(a_2, b_2) = -4(1-a_2)(1-b_2) - 5(1-a_2)b_2 - 2a_2b_2 = 4a_2 - 4 - b_2(1+a_2)$.

Obviously,

$$\max_A \$_1(A(0), b_2) \text{ (for any } b_2) \quad \text{is achieved when} \quad A(0) = 0, \quad \text{and}$$

[1] Let us recall a brief definition of Nash equilibrium: *a tuple of strategies is Nash equilibrium if no player can gain by a change of strategy as long as all the other players remain unchanged.*

$$\max_B \$_2\left(a_2, B\left(0\right)\right) \text{ (for any } a_2\text{)} \quad \text{is achieved when} \quad B\left(0\right) = 0.$$

In other words, when players independently maximize their payoffs (by choosing identity operators $A = B = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$), they both get

$$\$_1\left(0,0\right) = \$_2\left(0,0\right) = -4.$$

On the other hand, the Pareto optimal decision would be to choose exchange operators $A = B = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with payoffs

$$\$_1\left(1,1\right) = \$_2\left(1,1\right) = -2.$$

2.2 Quantum version

Jens Eisert, Martin Wilkens, and Maciej Lewenstein in [8, 7] investigated the quantization of non-zero-sum games and in particular Prisoner's Dilemma. The physical model proposed for quantum games assumes the following prerequisites:

- a system of multiple qubits in a given initial state (usually in an entangled one);
- two or more players, each of which is provided at least one qubit and the ability to act on these qubits;
- a device to measure the final state of the system, which determines the outcome for each of the players.

The proposed quantum game process is somewhat analogous to the classical one:

1. The game board consists of a two-qubit quantum system.
   Its initial state is $\psi_1 = |00\rangle$.

1_q Before all players' actions, an entangling operator

$$\widehat{J} \in \left\{ \exp\left(i\gamma \begin{pmatrix} 0001 \\ 0010 \\ 0100 \\ 1000 \end{pmatrix} \right) : \gamma \in \left[0, \frac{\pi}{4}\right] \right\}$$

   is applied to the system, which gets "entangled" into state $\psi_{1_q} = \widehat{J}\psi_1$.

2. Players independently choose operators $\widehat{A} = \widehat{U}\left(\theta_A, \phi_A\right)$ and $\widehat{B} = \widehat{U}\left(\theta_B, \phi_B\right)$ and apply them to the first and the second qubits of the system, respectively.
   Its state changes to $\psi_2 = \left(\widehat{A} \otimes \widehat{B}\right)\psi_{1_q}$.

2_q After all the players' actions are done, an operator $\widehat{J}^\dagger$ is applied to the system.
   Its state changes to $\psi_{2_q} = \widehat{J}^\dagger\psi_2$.
   Let us denote $\psi_{2_q} = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$.

3. The resulting qubits are measured in computational basis, and absolute squares of probability amplitudes $|\alpha_{ab}|^2$ are treated as probabilities of choosing strategy a by the first player and strategy b by the second player (where

$a, b \in \{0, 1\}$, strategy 0 corresponds to confessing and strategy 1 corresponds to denying participation in the crime).

4. Players get their payoffs — respectively
$\$_1\left(\psi_{2_q}\right) = -4\left|\alpha_{00}\right|^2 - 5\left|\alpha_{10}\right|^2 - 2\left|\alpha_{11}\right|^2$ and
$\$_2\left(\psi_{2_q}\right) = -4\left|\alpha_{00}\right|^2 - 5\left|\alpha_{01}\right|^2 - 2\left|\alpha_{11}\right|^2$

As one can see, the quantum scheme implies two additional steps 1_q and 2_q, which stand for the "quantum entangling" and "quantum disentangling" of quantum system, respectively.

The authors of [8] argue that (a) these two steps together with (b) two additional variables ϕ_A and ϕ_B in step 2, essentially make all the difference between the classical and quantum versions of the game. Therefore, the quantum version of the Prisoner's Dilemma is somewhat a: (a) generalization and (b) "quantum-mixed" expansion of the original game.

The fact is that by fixing $\phi_A = \phi_B = 0$ makes the game completely equivalent to its classical version with $a = \sin^2\theta_A$ and $b = \sin^2\theta_B$ (and the same Nash equilibrium $\theta_A = \theta_B = 0$ with payoffs $\$_1 = \$_2 = -4$) [2].

Therewith, adjusting parameter γ for operator $\widehat{J}$ from step 1_q allows obtaining different degrees of quantum entanglement effect in the game. Thus, setting $\gamma = 0$ eliminates any quantum entanglement and leaves the Nash equilibrium at the point of mutual confessing and a four-year prison sentence. On the other hand, setting $\gamma = \frac{\pi}{4}$ provides maximal entanglement, and improves the players' behavior up to the Pareto optimal mutual denying participation in the crime. For example, the set of Nash equilibria in such a game contains a symmetric pair of strategies $\widehat{A} = \widehat{B} = \widehat{U}\left(0, \frac{\pi}{4}\right)$ with payoffs $\$_1 = \$_2 = -2$. (The idea is the following: for the first player choosing strategy $\widehat{A} = \widehat{U}\left(0, \frac{\pi}{4}\right)$ guarantees equity $\left|\alpha_{01}\right|^2 = \left|\alpha_{10}\right|^2 = \frac{\sin^2\theta_B}{2}$, so that $\$_1\left(\psi_{2_q}\right) = \$_2\left(\psi_{2_q}\right)$, and with certainty converts an adversary into an ally; a similar argument remains for choosing strategy $\widehat{B} = \widehat{U}\left(0, \frac{\pi}{4}\right)$ by the second player.)

3 Technicalities of Game Quantization

3.1 Settings

Let us consider an arbitrary N-player game, where i^{th} player has arbitrary number (say M_i) of options (i.e. pure classical strategies).

We shall now treat this game in the spirit of the considerations applied to the Prisoner's Dilemma (see Section 2). Yet lifting of restrictions on the number of choices will lead to a different quantization method.

[2] This "backward compatibility" with the set of classical strategies is assured by the constraint on $\widehat{J}$ in step 1_q.

3.2 Classical formalism

We can formulate the settings of the game and the behavior of the players in terms of linear algebra in the following way:

1. The game board consists of N unit vectors, each from $\{0,1\}^{M_i}$ for $i \in \{1,2,\ldots,N\}$.
   Initially their values are $a_{1i} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$.

2. All players independently choose shift operators $A_1, A_2, \ldots, A_N$ and apply them one by one to $a_{11}, a_{12}, \ldots, a_{1N}$. Thus the ith player chooses $M_i \times M_i$ matrix

$$A_i \in \left\{ \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}^k = [m_{yx}] : m_{yx} = \begin{cases} 1 \text{ if } y+k \equiv x (\text{mod } N), \\ 0 \text{ otherwise} \end{cases}, k \in \mathbb{N} \right\} \tag{2}$$

   and multiplies it with a_{1i}.
   Let us denote $a_{2i} = A_i a_{1i}$ for $i \in \{1,2,\ldots,N\}$.

3. Each resulting vector a_{2i} is treated as probability distribution of choosing among M_i options available for the ith player.

4. Players get their payoffs according to the table of payoffs.

3.3 Quantum formalism

We now propose the general scheme for constructing a quantum version of an arbitrary miltiplayer multichoice game. This scheme is somewhat an analog and an expansion of the classical scheme described in Subsection 3.2.

1. The game board consists of an N-qudit quantum system, where the ith qudit is an M_i-level quantum subsystem.
   The initial state of the game board is $\psi_1 = |11\ldots1\rangle$ (and corresponds to choosing the first option by each of N players).

1_q Before all players' actions, an entangling operator $\widehat{J}$ (from the set, which we shall define later) is applied to the system, which gets "entangled" into state $\psi_{1_q} = \widehat{J}\psi_1$.

2. Players independently choose shift or any other unitary operators $\widehat{A}_1, \widehat{A}_2, \ldots, \widehat{A}_N$ and apply them one by one to the N qudits of the system. Its state changes to $\psi_2 = \left(\widehat{A}_1 \otimes \widehat{A}_2 \otimes \ldots \otimes \widehat{A}_N\right) \psi_{1_q}$.

2_q After all players' actions are done, an operator $\widehat{J}^\dagger$ is applied to the system. Its state changes to $\psi_{2_q} = \widehat{J}^\dagger \psi_2$.
Let us denote $\psi_{2_q} = \sum_{k_1=1}^{M_1} \sum_{k_2=1}^{M_2} \ldots \sum_{k_N=1}^{M_N} \alpha_{k_1 k_2 \ldots k_N} |k_1 k_2 \ldots k_N\rangle$.

3. Resulting qubits are measured in the computational basis, and the absolute squares of the probability amplitudes $|\alpha_{k_1 k_2 \ldots k_N}|^2$ (of total number $\prod_{i=1}^{N} M_i$) are treated as probabilities of simultaneous choosing options $k_1, k_2, \ldots, k_M$.

4. Players get their payoffs according to the table of payoffs.

In order to keep "backward compatibility" with the set of classical strategies (like in the Quantum Prisoner's Dilemma) we must apply special constraints on the entangling operator $\widehat{J}$ in step 2_q. Namely, we must ensure that steps 1_q and 2_q won't affect state ψ_{2_q} in the case where all players choose unitary operators from the set (2) which is "classical" subset of "quantum" set of strategies.

In terms of linear algebra this requirement is being defined as a set of $\prod_{i=1}^{N} M_i$ equalities:

$$\widehat{J}^\dagger \left(\bigotimes_{i=1}^{N} \widehat{A}_i \right) \widehat{J} = \bigotimes_{i=1}^{N} \widehat{A}_i \quad \text{(where all } \widehat{A}_i \text{ are from the set (2))},$$

which has $\left(\prod_{i=1}^{N} M_i\right)$-dimensional space of solutions:

$$\widehat{J} \in \left\{ \sum_{k_1=1}^{M_1} \sum_{k_2=1}^{M_2} \ldots \sum_{k_N=1}^{M_N} \lambda_{k_1 k_2 \ldots k_N} \left(\bigotimes_{j=1}^{N} E_{k_j} E_{k_j}^\dagger \right) : |\lambda_{k_1 k_2 \ldots k_N}| = 1 \right\},$$

where $E_{k_j} = \dfrac{1}{\sqrt{M_j}} \begin{pmatrix} \exp\left(2\pi i (k_j - 1) \frac{0}{M_j}\right) \\ \exp\left(2\pi i (k_j - 1) \frac{1}{M_j}\right) \\ \vdots \\ \exp\left(2\pi i (k_j - 1) \frac{M_j - 1}{M_j}\right) \end{pmatrix}$ is k_j^{th} eigenvector of $M_j \times M_j$ shift matrix.

4 Voting Game

4.1 Introduction

Voting games are used as a model for analyzing different political processes, and in particular elections.

When more than two candidates are participating, voters sometimes have reason for the so-called *tactical voting*. Namely, a voter in some situations doesn't support the most preferred candidate, and acts just in order to prevent the undesirable candidate from winning an election.

There is a large variety of voting systems, including ones with rather exotic ballots and tallying methods. In this section we shall consider the basic case of a plurality voting system, where this tactical voting occurs in a quantum model of the game and brings a better outcome to all the voters.

Let us consider elections with 4 candidates and only 3 voters. Voters give their votes to exactly one candidate each, and the single winner (the candidate with the most votes) provides the appropriate payoffs to all voters according to his personal preferences. If there are more than one candidate with maximal number of votes, let us pick one of the leaders at random and assert him to be the winner (in this case we shall consider voters' payoffs as expected values).

It is convenient to call all the participants with appropriate names and represent the voters' payoffs for all four possible winning candidates as in Table 2:

Table 2. Voters' payoffs

		Candidates			
		Mr. A	Mr. B	Mr. C	Mr. Fairness
Voters	Mrs. A	3	0	0	2
	Mrs. B	0	3	0	2
	Mrs. C	0	0	3	2

4.2 Nash equilibria

When analyzing *plurality* (or *relative majority*) voting it is worth to mention that not all Nash equilibria can be treated as solutions.

E.g. let us consider such voting game with N voters. Each situation when at least $\lfloor \frac{N}{2} \rfloor + 2$ voters vote for one (possibly, very bad) candidate, formally speaking, is a Nash equilibrium: there is no a single voter who could improve his outcome by switching to another option (since a single vote is unable to affect the absolute majority). Of course, unanimous voting for an unfit candidate could not be considered as a stable one or even as making any common sense. What is the appropriate concept of decision in such situations?

For different practical purposes the so-called *refinements* must be applied on the set of Nash equilibria. E.g. in real voting processes a voter usually

- is not sure about other voters' decisions, especially when tactical voting makes sense;
- is not sure whether or not all the voters will participate in voting;
- is not sure whether or not some voters by accident will vote for undesired candidates or just spoil their voting bulletins;
- is not sure about how many mistakes will be made during checking and summing up the results of the voting;

At least some of these factors must be taken into account. In terms of game theoretical analysis it means applying the so-called *trembling hand refinement*. *Trembling hand perfect equilibrium* assumes a small probability ϵ for each player to change his choice to some different one [14].

Thus in the voting game described above, *each* single vote has a chance to affect the majority, even if the probability of this effect is very small.

According to this refinement, let us assume that after voters completed their task, the tallying machine counts each voting bulletin properly with probability $1 - 3\epsilon$, and with probability 3ϵ makes mistake (in favor to one of the remaining candidates, with probability ϵ for each candidate).

Then the probabilities of winning in different cases of voting are as follows:

Table 3. Adjustment of winning probabilities for trembling hand perfect equilibrium

<table>
<tr><th>Combinations of votes</th><th>Pr [Mr. A wins]</th><th>Pr [Mr. B wins]</th><th>Pr [Mr. C wins]</th><th>Pr [Mr. Fairness wins]</th></tr>
<tr><td>3 × Mr. A</td><td>$1 - 21\epsilon^2 + 36\epsilon^3$</td><td colspan="3">$7\epsilon^2 - 12\epsilon^3$</td></tr>
<tr><td>2 × Mr. A,
1 × Mr. B</td><td>$1 - 4\frac{2}{3}\epsilon + 9\frac{2}{3}\epsilon^2 - 12\epsilon^3$</td><td>$3\frac{1}{3}\epsilon - 13\epsilon^2 + 14\frac{2}{3}\epsilon^3$</td><td colspan="2">$\frac{2}{3}\epsilon + 1\frac{2}{3}\epsilon^2 - 1\frac{1}{3}\epsilon^3$</td></tr>
<tr><td>1 × Mr. A,
1 × Mr. B,
1 × Mr. C</td><td colspan="3">$\frac{1}{3} - \frac{1}{3}\epsilon + \frac{1}{3}\epsilon^2 - 1\frac{1}{3}\epsilon^3$</td><td>$\epsilon - \epsilon^2 + 4\epsilon^3$</td></tr>
</table>

All other cases are equivalent to above-stated ones up to permutation of candidates.

4.3 Classical version

Let us analyze the voting game with payoffs defined in Table 2.

The single trembling hand perfect equilibrium is "selfish voting": it is easy to show that regardless of the other votes, Mrs. A's best choice is to vote for Mr. A. Similar arguments will show that Mrs. B will vote for Mr. B, and Mrs. C will vote for Mr. C.

Mr. Fairness, whose leadership promises the biggest common benefit, will remain with no votes at all.

Summing up these arguments, we follow that Messrs. A, B and C will win with probabilities about $\frac{1}{3}$ each, thus providing Mesdames A, B and C with benefits about 1 for each of them [3], [4].

Let us now reformulate the process of voting game in terms of linear algebra, like it was done in Section 2 for Prisoner's Dilemma:

1. The game board consists of three unit vectors from $\{0,1\}^4$.
   Initially their values are $a_1 = b_1 = c_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$.

2. Voters independently choose shift operators A, B, C from $\left\{ \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}^k : k \in \mathbb{N} \right\}$ and apply them respectively to a_1, b_1, c_1.
   Let us denote $a_2 = Aa_1, b_2 = Bb_1, c_2 = Cc_1$.

3. Resulting vectors a_2, b_2, c_2 are treated as probability distribution of choosing among four options available for voters.
   Note that the set of allowed operators in step 2 implement only pure strategies for voters. This restriction however is not crucial as in voting games use of mixed strategies generally makes no sense.

4. Winner is selected among the candidates according to Table 3, and voters get their payoffs according to Table 2.

[3] To be completely precise in accordance to Table 3, each voter's payoff is $1+\epsilon-\epsilon^2+4\epsilon^3$, i.e. slightly bigger, because Nash equilibrium here is the worst decision in terms of common benefit, and deviation from this point (caused by *trembling hand*) improves the payoffs.

[4] Generally speaking, the set of Nash equilibria in such voting games might not consist of a single element (as it does in our case), but all Nash equilibria assume voters to use pure strategies (unlike general bimatrix or multimatrix games, where use of mixed strategies sometimes prevent players from big loss).

Thus described above Nash equilibrium of our voting game asserts the following values:

$$a_1 = b_1 = c_1 = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}$$

$$A = \begin{pmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{pmatrix}^0 = \begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}, \quad B = \begin{pmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{pmatrix}^1 = \begin{pmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{pmatrix},$$

$$C = \begin{pmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{pmatrix}^2 = \begin{pmatrix}0&0&1&0\\0&0&0&1\\1&0&0&0\\0&1&0&0\end{pmatrix}$$

$$a_2 = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \qquad b_2 = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \qquad c_2 = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}$$

4.4 Quantum version

We propose a quantization of this voting game by the general method described in Subsection 3.3. In the following voting scheme we numbered candidates from 0 to 3 as they appear in Table 2.

1. The game board consists of a three-qudit quantum system, where each qudit is a 4-level quantum subsystem.
   Initial state of the game board is $\psi_1 = |000\rangle$.

1_q Before all players' actions, an entangling operator $\widehat{J}$ (from the set, which we shall define later) is applied to the system, which gets "entangled" into state $\psi_{1_q} = \widehat{J}\psi_1$.

2. Players independently choose unitary operators $\widehat{A}, \widehat{B}, \widehat{C}$ and apply them one by one to the three qudits of the system. Its state changes to $\psi_2 = \left(\widehat{A} \otimes \widehat{B} \otimes \widehat{C}\right)\psi_{1_q}$.

2_q After all players' actions are done, an operator $\widehat{J}^\dagger$ is applied to the system. Its state changes to $\psi_{2_q} = \widehat{J}^\dagger\psi_2$.
   Let us denote $\psi_{2_q} = \sum_{i=0}^{3}\sum_{j=0}^{3}\sum_{k=0}^{3}\alpha_{ijk}|ijk\rangle$.

3. Resulting qubits are measured in computational basis, and the absolute squares of amplitudes $|\alpha_{ijk}|^2$ are treated as probabilities of voting for candidates i, j, k by the first, second and third voters respectively.

4. Winner is selected among the candidates according to Table 3, and voters get their payoffs according to Table 2.

In order to keep "backward compatibility" with the set of classical strategies (like in the Prisoner's Dilemma) we must apply special constraints on the entangling operator $\widehat{J}$ in step 2_q. Namely, we must ensure that steps 1_q and 2_q won't affect state ψ_{2_q} in the case where all three voters choose unitary operators from the set $\left\{ \begin{pmatrix} 0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0 \end{pmatrix}^k : k \in \mathbb{N} \right\}$, so that they are equivalent to shift operators from the classical version of game.

In terms of linear algebra this requirement is being defined as a set of $4^3 = 64$ equalities:

$$\widehat{J}^\dagger \left(\widehat{A} \otimes \widehat{B} \otimes \widehat{C} \right) \widehat{J} = \widehat{A} \otimes \widehat{B} \otimes \widehat{C}, \quad \text{where } \widehat{A}, \widehat{B}, \widehat{C} \in \left\{ \begin{pmatrix} 0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0 \end{pmatrix}^k : k \in \mathbb{N} \right\},$$

which has 64-dimensional space of solutions:

$$\widehat{J} \in \left\{ \sum_{i=0}^{3} \sum_{j=0}^{3} \sum_{k=0}^{3} \lambda_{ijk} \left(E_i E_i^\dagger \otimes E_j E_j^\dagger \otimes E_k E_k^\dagger \right) : |\lambda_{ijk}| = 1 \right\},$$

where $E_0 = \frac{1}{2} \begin{pmatrix} 1\\1\\1\\1 \end{pmatrix}, E_1 = \frac{1}{2} \begin{pmatrix} 1\\i\\-1\\-i \end{pmatrix}, E_2 = \frac{1}{2} \begin{pmatrix} 1\\-1\\1\\-1 \end{pmatrix}, E_3 = \frac{1}{2} \begin{pmatrix} 1\\-i\\-1\\i \end{pmatrix}$ are eigenvectors of 4×4 shift matrices.

As an example we shall consider a symmetric bundle of λ's:

$$\lambda_{ijk} = \begin{cases} 1 & \text{if } i = j = k, \\ -1 & \text{otherwise,} \end{cases}$$

and corresponding 64×64 matrix

$$\widehat{J} = [m_{ij}], \text{ where } m_{ij} = \begin{cases} -\frac{7}{8} & \text{if } i = j, \\ \frac{1}{8} & \text{if } i + \lfloor \frac{i}{4} \rfloor + \lfloor \frac{i}{16} \rfloor \equiv j + \lfloor \frac{j}{4} \rfloor + \lfloor \frac{j}{16} \rfloor \pmod 4, \\ 0 & \text{otherwise.} \end{cases}$$

4.5 Search for quantum Nash equilibrium

Although Nash equilibrium is known as the most important solution concept of non-zero-sum games, very little is known about efficient computing of Nash equilibria for an arbitrary game. For example, Megiddo and Papadimitriou [12] placed this problem for a *finite two-player game* into rather specific complexity class TFNP, and question about the precise computational complexity is still relevant.

Certainly, our problem for *infinite multiplayer game* is much less clear. In this section we shall only (a) show inconsistency of classical selfish voting in terms of Nash equilibrium and (b) provide a promising solution, which turns up, in a curious way, Pareto optimal.

All our considerations are based on numerical experiments, which model the process of iterative search for equilibrium point in the multidimensional space of strategies.

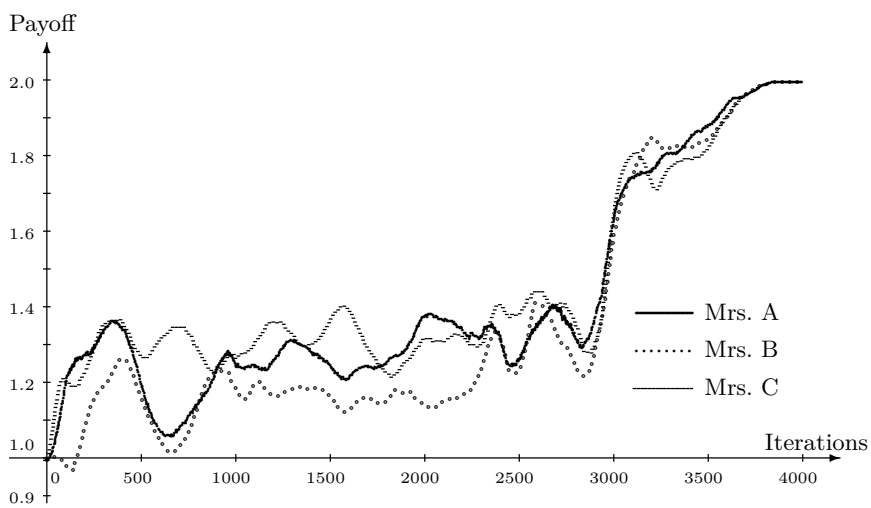
We start our search from an arbitrary point. So as to show inconsistency of classical Nash equilibrium, let us start with the selfish voting, i.e. with

$$\widehat{A} = \begin{pmatrix} 1&0&0&0\\ 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1 \end{pmatrix}, \qquad \widehat{B} = \begin{pmatrix} 0&1&0&0\\ 0&0&1&0\\ 0&0&0&1\\ 1&0&0&0 \end{pmatrix}, \qquad \widehat{C} = \begin{pmatrix} 0&0&1&0\\ 0&0&0&1\\ 1&0&0&0\\ 0&1&0&0 \end{pmatrix}.$$

Then let us perform a series of the iterations. We first let Mrs. A slightly [5] change her unitary operator $\widehat{A}$ in order to increase her expected payoff (assuming $\widehat{B}$ and $\widehat{C}$ to remain unchanged). Then we make similar offer to Mrs. B, and then to Mrs. C. After that we repeat these three adjustments, starting again with Mrs. A, until some stable point is reached.

Changes of voters' payoffs during this iterative algorithm is displayed in Figure 1.

Fig. 1. Voters' payoffs during evolution



Sooner or later, voters' payoffs reach the neighborhood of a Pareto optimal point (and never leave it again), where each voter has payoff 2.

Basically, results of this (probably not the most efficient) algorithm suffice for both our purposes:

[5] Say we can restrict $\max_{i,j} \left| \widehat{A}_{ij} - \widehat{A}'_{ij} \right| \leq 0.01$, where $\widehat{A}'$ is Mrs. A's operator after the change is made. Actually we just restricted the distance between $\widehat{A}$ and $\widehat{A}'$ in the parametrized space of $M \times M$ unitary operators.

(a) It shows inconsistency of classical selfish voting in terms of Nash equilibrium. We can follow that if Mrs. B and Mrs. C stick to their classical optimal strategies, i.e. $\widehat{B} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$ and $\widehat{C} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$, then there exists a strategy for Mrs. A, which is significantly better than $\widehat{A} = I$. E.g. choosing $\widehat{A} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ will provide her with expected outcome $\frac{101}{64}$ (while Mrs. B will get $\frac{47}{64}$, and Mrs. C will get $\frac{53}{64}$).

(b) It provides us with a non-rigorous equilibrium, which probably could be proven analytically. We shall not go into greater detail, as this was just a specific example for illustration of the general game quantization method.

5 Possible Issues

5.1 Objections on EWL scheme

In [6] van Enk and Pike provide some criticism on the quantum Prisoner's Dilemma constructed by Eisert et al. in [8, 7]. Namely, they argue that this generalization of classical Prisoner's Dilemma is somewhat equivalent to offering a new classical choice $\boldsymbol{Q}$ for each of the players, so that the payoff matrix in Table 1 is expanded as in Table 4.

Table 4. Players' payoffs in Prisoner's Dilemma with an extra choice $\boldsymbol{Q}$

		Your partner		
		Confesses	Denies	$\boldsymbol{Q}$s
You	Confess	4y. for You, 4y. for Your partner	0y. for You, 5y. for Your partner	**5y.** for You, **0y.** for Your partner
	Deny	5y. for You, 0y. for Your partner	2y. for You, 2y. for Your partner	**4y.** for You, **4y.** for Your partner
	$\boldsymbol{Q}$	**0y.** for You, **5y.** for Your partner	**4y.** for You, **4y.** for Your partner	**2y.** for You, **2y.** for Your partner

Obviously, a mutual choice of $\boldsymbol{Q}$ is the single Nash equilibrium.

Additionally, authors of [6] provide some arguments of rather philosophical nature. Namely, they treat step 2_q as an activity of an attorney or interroga-

tor which actually helps the prisoners. Providing prisoners with such assistance certainly is counter to the spirit of the game.

Finally, Benjamin and Hayden [2] showed that when the set of strategies is not restricted with two-parametric $\widehat{U}(\theta, \phi)$, and players are allowed to use any local unitary operator, then strategies $\widehat{A} = \widehat{B} = \widehat{U}\left(0, \frac{\pi}{4}\right)$ do not remain a Nash equilibrium. Moreover, they show that in such settings no Nash equilibrium for the quantum Prisoners' Dilemma exists.

This criticism, of course, should be taken into account when constructing a quantum version of a game. Although first two objections (by van Enk and Pike) seem not to be that relevant for the voting game (described in Section 4), the comment by Benjamin and Hayden is worth some complementary discussion.

5.2 Comment on Nash equilibria in quantized games

The fact is that quantization by [8] is claimed to be a generalization of the *mixed expansion* of Prisoner's Dilemma. But, as it is noted in [2], Nash's theorem may not work any more under these circumstances.

Unlike quantization of Prisoner's Dilemma with exactly two pure classical strategies per player, the more general quantization proposed in this part is claimed to be an expansion, which does not implement mixed strategies in the sense of classical probabilities [6]. It is not clear whether Nash's theorem works for this *quantum expansion* of the game. While it might not be relevant for the quantum voting game, overall situation needs some futher reasoning on existance of Nash equilibrium points.

For both quantization schemes (by [8, 7] and by one proposed in this work) the following reasoning seems to be quite natural. Since sets of quantum strategies are compact, and the payoff function is a continuous mapping to the compact (N-dimensional) space of payoffs, one can observe such quantum game as a basic case of classical game with infinite set of strategies. In game theory it is a well-known fact that for mixed expansion of such game exists a Nash equilibrium [9]. In particular, for quantum Prisoner's Dilemma all Nash equilibrium points consist of mixed strategies. In general, search for a Nash equilibrium may require analysis of superoperators as players' strategies and mixed quantum states as objects for the final measurement of step 2_q.

From this point of view the only fundamental difference between the two quantization schemata is as follows:

(a) EWL scheme is a "quantum-over-mixed" expansion of an original classical game;
(b) our scheme is just a "quantum" expansion, which contains only *pure strategies* as a special case, and does not contain *mixed strategies* in the sense of classical probabilistic choice.

[6] This feature can be qualified as a drawback in comparison with EWL scheme, but our further considerations in this section show that it is not that significant. In fact, implementing mixed strategies appear to be obsolete, when speaking about Nash equilibria.

But in both cases, generally speaking, we need to apply further mixed expansion (one can call it "*mixed-over-quantum expansion*") of the resulting game in order to figure out its Nash equilibria.

Summary

The strategies of the games with quantum decisions are operations undertaken on qubits, whilst players' winnings are calculated taking into account the results of quantum system measurements. The classical choices of players account only for a limited ratio of various quantum operations, which often does not allow the maximally effective approach by rationally thinking players. The usage of quantum strategies makes it possible to sufficiently enlarge players' abilities to interact among themselves applying quite rational principles.

Quantum Prisoner's Dilemma (see Section 2 may serve as a basic illustration of a game with quantum decisions. In Section 3 we provided a technique, which is of similar spirit but technically different, which allows quantization of games with arbitrary many players and arbitrary many players' choices.

In Section 4 we illustrated ideas of Section 3 by quantizing a specific noncooperative game, which models the process of elections. We also provided computer experiments modeling the behavior of participants in a system, which is working under the laws of quantum theory.

In Section 5 we study some objections on quantum Prisoner's Dilemma which arose after work [8]. We analyzed these issues (by Benjamin and Hayden [2], and by van Enk and Pike [6]), and evaluated their relevance to our proposed quantization scheme of Section 3 and to the specific quantized voting game of Section 4. We followed that although the above-mentioned objections may be not that relevant to the voting game, some of them must be born in mind when constructing new quantum versions of classical games.

Conclusion

Quantum games are of particular interest due to the fact that they can already be implemented in physical experiments The fact is that implementing most of these games only requires operations on separate qubits, and all the necessary technologies are already available in experimental quantum optics. For example, Chinese physicists successfully conducted a number of physical experiments [5] based on the theoretical model described in [8]; thus, confirming the general applicability of the EWL scheme.

Therefore, one can say that, despite its relative youth, the theory of quantum games is already confirmed experimentally. This fact distinguishes quantum games from other areas of quantum computing, which often require complex technologies to manipulate large quantum systems, and have not yet been confirmed in large-scale experiments.

On the other hand, studies of some theoretical aspects of quantum games remain insufficient.

Thus games with quantum decisions have so far been presented only as analogs of classic games with binary strategies. (Probably the only exception is the recently introduced quantum Kolkata Restaurant Problem [15] with three strategies for each player.) This work makes an effort to fill this gap, but it can only be considered as a first step on the way to studying quantum games with an arbitrary number of strategies.

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